

## HALD VEST, Denmark

WMO: 060490

Lat: 56.561N

Lon: 10.093E

Elev: 89

StdP: 100.26

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -7.7       | -6.0 | -10.5                         | 1.5 | -6.2 | -8.8 | 1.8 | -4.6 | 20.2                  | 5.0  | 16.3 | 4.9  | 3.7                   | 350  | 0.683 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.7                    | 25.6            | 18.3 | 23.8 | 17.5 | 22.1 | 16.7 | 19.3                | 23.5 | 18.3 | 22.2 | 17.4 | 21.0 | 4.6                  | 80   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 17.7                            | 12.8 | 20.7 | 16.8 | 12.1 | 19.9 | 16.0 | 11.5 | 19.2 | 55.2          | 23.6 | 52.1 | 22.2 | 49.3 | 21.0 | 22.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 14.2          | 11.7 | 10.2 | -9.1                       | 28.6 | 2.1                | 1.6 | -10.5                                              | 29.7 | -11.8      | 30.7 | -12.9      | 31.6 | -14.4      | 32.8 |    |    |
| (5)               |      |      | -9.8                       | 20.9 | 2.4                | 0.9 | -11.5                                              | 21.5 | -13.0      | 22.0 | -14.3      | 22.5 | -16.1      | 23.1 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 8.3    | 0.6   | 0.8   | 3.1   | 7.2   | 11.1  | 13.9  | 16.5  | 16.5  | 13.4  | 9.1   | 5.0   | 2.2  |
|                                                                                  | DBStd       | 6.50   | 3.61  | 3.48  | 3.37  | 3.18  | 3.23  | 2.70  | 2.82  | 2.51  | 2.53  | 2.77  | 2.99  | 3.70 |
|                                                                                  | HDD10.0     | 1332   | 292   | 258   | 213   | 95    | 26    | 2     | 0     | 0     | 2     | 49    | 152   | 242  |
|                                                                                  | HDD18.3     | 3689   | 550   | 492   | 471   | 334   | 226   | 137   | 72    | 70    | 151   | 285   | 401   | 501  |
|                                                                                  | CDD10.0     | 717    | 0     | 0     | 1     | 11    | 59    | 117   | 203   | 201   | 103   | 23    | 1     | 0    |
|                                                                                  | CDD18.3     | 33     | 0     | 0     | 0     | 0     | 1     | 3     | 17    | 12    | 1     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 196    | 0     | 0     | 0     | 1     | 1     | 20    | 112   | 59    | 3     | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 19     | 0     | 0     | 0     | 0     | 0     | 1     | 12    | 6     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 5.1    | 5.9   | 5.8   | 5.7   | 5.1   | 4.7   | 4.6   | 4.0   | 4.3   | 4.8   | 5.2   | 5.5   | 5.5  |
| Precipitation                                                                    | PrecAvg     | 702    | 59    | 43    | 45    | 41    | 50    | 62    | 63    | 75    | 67    | 74    | 64    | 60   |
|                                                                                  | PrecMax     | 892    | 106   | 118   | 97    | 95    | 96    | 134   | 117   | 167   | 189   | 140   | 127   | 123  |
|                                                                                  | PrecMin     | 496    | 6     | 10    | 5     | 6     | 8     | 2     | 14    | 23    | 15    | 16    | 24    | 21   |
|                                                                                  | PrecStd     | 100    | 27    | 23    | 24    | 25    | 20    | 27    | 29    | 35    | 38    | 30    | 26    | 28   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 9.2   | 9.7   | 13.1  | 18.9  | 22.8  | 25.9  | 28.4  | 27.5  | 23.4  | 17.0  | 12.8  | 10.6 |
|                                                                                  |             | MCWB   | 8.3   | 8.4   | 8.9   | 12.9  | 16.5  | 18.3  | 19.0  | 18.7  | 16.8  | 14.5  | 11.6  | 9.5  |
|                                                                                  | 2%          | DB     | 7.7   | 7.8   | 10.7  | 16.3  | 20.7  | 23.2  | 26.1  | 24.9  | 21.1  | 15.2  | 11.4  | 9.2  |
|                                                                                  |             | MCWB   | 7.0   | 6.7   | 7.9   | 11.1  | 15.4  | 17.2  | 18.4  | 18.0  | 16.2  | 13.4  | 10.5  | 8.5  |
|                                                                                  | 5%          | DB     | 6.7   | 6.7   | 8.9   | 14.4  | 18.6  | 21.2  | 24.5  | 23.1  | 19.0  | 14.0  | 10.2  | 8.1  |
|                                                                                  |             | MCWB   | 5.9   | 5.8   | 6.6   | 9.9   | 13.9  | 16.3  | 18.0  | 17.3  | 15.0  | 12.6  | 9.4   | 7.5  |
|                                                                                  | 10%         | DB     | 5.7   | 5.5   | 7.5   | 12.4  | 16.9  | 19.2  | 22.4  | 21.2  | 17.4  | 13.0  | 9.2   | 7.2  |
|                                                                                  |             | MCWB   | 5.0   | 4.8   | 5.8   | 9.0   | 13.1  | 15.0  | 17.1  | 16.2  | 14.3  | 11.9  | 8.5   | 6.6  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 8.6   | 8.5   | 9.9   | 13.9  | 17.8  | 19.5  | 20.6  | 20.1  | 17.9  | 15.0  | 11.9  | 9.7  |
|                                                                                  |             | MCDB   | 9.1   | 9.3   | 12.2  | 17.4  | 21.7  | 23.7  | 25.5  | 24.2  | 21.3  | 16.3  | 12.6  | 10.2 |
|                                                                                  | 2%          | WB     | 7.1   | 7.0   | 8.2   | 11.8  | 15.9  | 18.0  | 19.6  | 18.9  | 16.6  | 13.8  | 10.6  | 8.6  |
|                                                                                  |             | MCDB   | 7.6   | 7.5   | 10.2  | 14.9  | 19.1  | 22.2  | 24.2  | 23.0  | 19.9  | 14.7  | 11.3  | 9.0  |
|                                                                                  | 5%          | WB     | 6.1   | 5.8   | 7.0   | 10.4  | 14.7  | 16.7  | 18.6  | 18.0  | 15.7  | 12.8  | 9.6   | 7.6  |
|                                                                                  |             | MCDB   | 6.5   | 6.5   | 8.4   | 13.3  | 17.9  | 20.3  | 22.7  | 21.6  | 18.5  | 13.7  | 10.1  | 8.0  |
|                                                                                  | 10%         | WB     | 5.0   | 4.8   | 6.1   | 9.3   | 13.5  | 15.4  | 17.6  | 17.1  | 14.8  | 12.0  | 8.6   | 6.6  |
|                                                                                  |             | MCDB   | 5.6   | 5.5   | 7.3   | 12.0  | 16.3  | 18.4  | 21.3  | 20.0  | 16.8  | 12.9  | 9.2   | 7.1  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 3.7   | 3.8   | 5.4   | 7.2   | 8.1   | 8.3   | 8.7   | 8.1   | 6.8   | 4.8   | 4.0   | 3.6  |
|                                                                                  |             | MCDBR  | 4.3   | 4.7   | 7.8   | 9.8   | 10.3  | 10.9  | 11.2  | 10.3  | 8.7   | 5.6   | 4.6   | 4.1  |
|                                                                                  |             | MCWBR  | 4.3   | 4.0   | 5.3   | 5.9   | 6.4   | 6.4   | 5.2   | 4.6   | 4.6   | 4.1   | 4.3   | 4.1  |
|                                                                                  | 5% WB       | MCDBR  | 4.3   | 4.5   | 7.2   | 9.0   | 9.3   | 10.0  | 10.1  | 8.9   | 7.8   | 5.3   | 4.4   | 4.1  |
|                                                                                  |             | MCWBR  | 4.3   | 4.0   | 5.2   | 5.8   | 6.3   | 6.5   | 5.4   | 4.7   | 4.7   | 4.2   | 4.2   | 4.0  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.307  | 0.330 | 0.349 | 0.385 | 0.378 | 0.370 | 0.389 | 0.390 | 0.364 | 0.345 | 0.323 | 0.292 |      |
|                                                                                  | taud        | 2.241  | 2.296 | 2.321 | 2.277 | 2.343 | 2.388 | 2.356 | 2.365 | 2.424 | 2.409 | 2.303 | 2.218 |      |
|                                                                                  | Ebn at Noon | 572    | 721   | 807   | 829   | 860   | 872   | 844   | 817   | 787   | 699   | 550   | 483   |      |
|                                                                                  | Edh at Noon | 54     | 76    | 94    | 114   | 114   | 110   | 112   | 104   | 85    | 68    | 51    | 42    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.42   | 0.99  | 2.40  | 4.03  | 5.31  | 5.75  | 5.29  | 4.17  | 2.88  | 1.42  | 0.54  | 0.30  |      |
|                                                                                  | RadStd      | 0.08   | 0.17  | 0.22  | 0.47  | 0.48  | 0.45  | 0.56  | 0.33  | 0.29  | 0.18  | 0.09  | 0.04  |      |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (55 neighbors) |  | +0.38 | N/A     | N/A    | N/A     | N/A   | N/A   | -118        | -164    | N/A     | N/A     |

Nomenclature: See separate page